

Reducing Fragmentation in Digital Health: An HCI-Driven Design of a Centralized Healthcare Platform

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0. Abstract

Reducing fragmentation and centralizing digital healthcare systems helps simplify user processes. This result is supported by research in human-computer interactions for medical applications and our own market research. Our requirement-gathering process included surveys of patients and interviews with medical professionals, and revealed widespread frustration with current fragmented healthcare applications. Building on the user requirements gathered, we developed an interactive medium-fidelity prototype to support primary care interactions for both patients and clinicians. Our usability study focuses on our two major flows (patients and healthcare workers). Using tools like Google Forms to conduct our questionnaire, the usability results indicated high task success and strong overall design clarity. However, it also showed issues related to button reliability and feedback consistency. These findings highlight both the effective parts of the interface and the parts that need refinement. The limitations look at how users interacted with our healthcare app prototype and highlight the main limitations we faced during the testing. Based on these insights, we outline practical future improvements, ranging from quick usability fixes to longer-term enhancements.

1. Introduction

Digital health systems have become central to modern care delivery, yet many remain fragmented across disconnected portals, record systems, and communication platforms. Patients frequently navigate multiple websites, phone systems, and hospital-specific portals to access basic care services such as booking appointments, retrieving lab results, or reviewing prescriptions. Clinicians face similar challenges, often working across incompatible

electronic health record (EHR) systems that disrupt workflow efficiency, hinder communication, and introduce risk for delayed or incorrect treatment. Phase I of this project identified fragmentation rather than lack of technology as the core barrier to effective digital health experiences for both clinicians and patients. Fragmentation reduces efficiency, increases cognitive load, reinforces inequalities in access, and ultimately compromises care quality.

This project discusses the design of a single, accessible, patient-centred digital health application that would support users through streamlined access to appointments, records, and prescriptions. By situating design in user-centred research-surveys, interviews, personas, and scenarios-and linking it with the current HCI and health-informatics literature, we aim at an understanding of how a one-stop platform can reduce fragmentation, enhancing usability, accessibility, and continuity of care.

2. Literature Review

Past research underlines general usability issues and systemic fragmentation of digital health platforms [1]. Demonstrate that even high-consequence public health systems, such as state COVID-19 vaccine sites, feature basic accessibility issues in the form of limited compatibility with smartphones, low readability, absence of updating indicators, and multi-step processes for key tasks such as appointment setting. These findings support the notion that poor usability is indicative not of a stand-alone problem but rather of a systemic symptom of digital infrastructures that are poorly integrated, which correlates with observations from Phase I concerning inefficiency, inconsistency, and inequitable access to care.

Fragmentation, on the clinician's side, is equally well-documented [2]. Report that the organizational and technological barriers of patient-accessible EHRs faced by clinicians include inconsistent templates, poor documentation by users, inaccessible systems for low-literacy users, and increased workload due to poor interface design. Their findings mirror the concerns in our Phase II interviews, where the medical trainees reported difficult retrieval of complete information about the patient, reliance on ad-hoc tools, such as screenshots and shared spreadsheets, and the burden of switching between multiple portals [6]. continue to highlight that EHRs support collaboration only when they provide consistent and shared overviews; otherwise, authorization barriers, information overload, and non-standard interfaces reduce communication efficiency and trust. These studies collectively reinforce the need for a unified, streamlined system in support of collaborative care.

Benefits of Integrated, Patient-Centred Digital Health Systems

On the patient side, literature has shown that unified access to health information improves outcomes, engagement, and satisfaction [5]. Find that EHRs accessible by patients increase medication adherence, reduce anxiety, and strengthen doctor-patient communication, with consideration of the risk of inequity when systems lack usability or accessibility features [4]. Similarly, show that patient-centred digital records improve self-management, treatment adherence, and health literacy.

Poor design serves to exclude the very populations that stand most to benefit from digital health tools [3]. Report that disabled users experience significantly higher usability barriers in terms of poor compatibility with assistive technologies, inaccessible interaction patterns, and a lack of inclusive design practices. This brings us directly to the Phase I requirements: high-contrast interfaces, text that can be adjusted, simplified, voice, and screen readers. As our survey data below highlights, accessibility features like adjustability in text size, biometric login, and screen-reader compatibility ranked among the most demanded features.

Design opportunities through user research

insights from Phase II interviews and surveys reinforce the literature's findings. Patients described challenges in appointment scheduling across different systems, delays in accessing results, and inconsistent portals. The clinicians and medical students highlighted centralization, ease of use, and role-based access as key characteristics for clinical efficiency. Almost all clinicians interviewed asked for a common system with unified patient records, quick-search functionality, smoother data flow, and visual summaries to assist with interpretation and decision-making. These highlights suggest that it is possible to design a platform that will fill three significant gaps:

1. Integration: Appointments, test results, prescriptions, and visit summaries can all be combined on this platform.
2. Accessibility: Usability for all abilities, devices, and contexts.
3. Role-differentiated design: Supporting the efficiency needs of doctors and comprehension needs of patients without compromising security or privacy.

3. Problem Statement & Research Question

3.1 Problem Statement

Current digital health systems are fragmented across disparate portals, isolated EHR systems, and inconsistent usability standards. Fragmentation poses significant barriers to both patients' and clinicians' activities, wherein the former face problems in accessibility and interpretation of health information, while the latter confront workflow disruptions and the inability to retrieve comprehensive records. Most current systems do not consider accessibility and provide a non-fragmented, seamless experience. As revealed by literature and Phase I–III findings, fragmentation decreases efficiency, coordination of care, usability, and equity.

3.2 Research Question

How might we design a single, accessible, digital health application that addresses the fragmentation of existing healthcare systems and supports both patients and clinicians in managing appointments, health records, and prescriptions more effectively?

Sub-questions that support this question include:

What are the critical usability and accessibility requirements identified by patients and clinicians for managing their health information?

How does system fragmentation impact efficiency, comprehension, and care coordination for disparate user groups?

Which design features and interaction patterns most effectively support a single, coherent workflow for Appointments, Records, and Prescriptions?

4. Gathering of User Requirements

4.1 Requirement Gathering

We adopted a user-centred design approach to ensure our healthcare application is designed around actual user needs. To understand the tasks, goals, and concerns of users within existing healthcare systems, we decided to use surveys and semi-structured interviews to gather requirements.

Early on, we focused on users and tasks, closely examining behavioural, cognitive, anthropomorphic, and attitudinal characteristics. For example, through our surveys, we explored patients' cognitive inclinations by asking how they interpret health information, how they recall appointment details, and whether they prefer visual accessibility features like flexible text or dark mode.

We surveyed 29 potential patients who would be using our app. Surveys are easy and quick to distribute, with a fast response time. Data can also be easily collected and analyzed in the Google Forms database with charts and graphs.

The 11 interviews focused on gathering in-depth qualitative insights from medical professionals, as they are the primary users responsible for maintaining health information. Semi-structured interviews provide more accurate and high-quality responses while allowing flexible probing into

complex issues, giving us a good balance between richness and replicability.

4.2 Key Findings

The survey revealed that most participants still rely on phone calls for appointments and email to track records. The major issues included difficulty finding available slots, long wait times, and managing multiple portals. These findings indicated that users lacked a unified, efficient tool for basic healthcare tasks. Participants also expressed a strong preference for biometric login, simple authentication, and accessibility features such as adjustable text size and high-contrast mode.

These findings indicated that users need a centralized, easy-to-navigate system with stronger accessibility and security support.

Our interviews highlighted that almost all medical professionals struggled to see complete patient histories across labs, clinics, and record systems. Many clinicians relied on handwritten notes or messaging apps due to poor existing tools. They consistently emphasized the need for a unified record system, visual analytics for lab trends, secure communication with patients, simple mobile interfaces, and role-based privacy controls to maintain ethical data access.

Overall, our user research revealed that both patients and clinicians are burdened by fragmented systems that require them to switch between multiple portals, communication channels, and record sources. Patients struggle with basic tasks such as booking appointments or viewing records, while medical professionals face even greater challenges when trying to access complete, accurate patient histories.

Across our requirement gathering methods, users expressed a desire for simplicity, consistency, and reliability. From these results, we concluded that an effective solution must unify appointments, records, and communication into a single coherent workflow while ensuring accessibility, security, and ease of use. These insights formed the foundation for the design directions pursued in the next phase of the project.

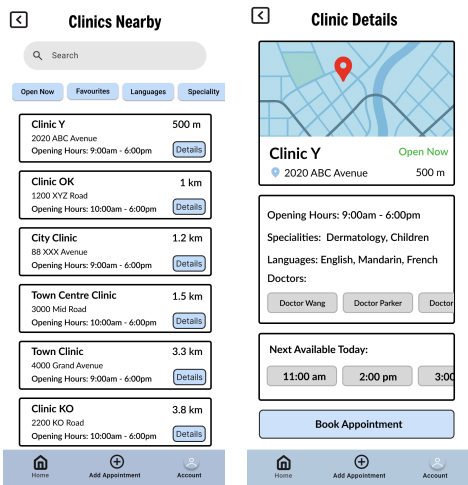
5. Prototype

5.1 Prototype Overview:

Building on the user requirements gathered in earlier phases, we developed an interactive medium-fidelity prototype to support primary care interactions for both patients and clinicians. The prototype was implemented in Figma with two major workflows: a patient-facing mobile application and a doctor-facing web dashboard. Together, these flows were designed to support three core tasks derived from the requirements analysis: (1) Nearby Clinic Discovery, (2) Appointments Booking and Management, and (3) Health Records and Prescriptions Management.

5.2 Task 1: Nearby Clinic Discovery (Patients)

Task 1 enables patients to efficiently discover and locate suitable clinics in their area. From the patient’s home screen, users can click “Check Clinics Nearby” to open the “Clinics Nearby” page (Fig 5.2a) showing a scrollable list of clinics automatically sorted by distance. Each item presents high-level information (e.g. name, opening hours, distance, address) to support quick scanning. The search bar and filters (e.g. specialty, languages) allow patients to further narrow down options. Clicking the “Details” button opens a clinic profile with richer information (Fig 5.2b), including an embedded map, specialties, available doctors, and upcoming appointment availability. This flow supports patients in identifying an appropriate nearby clinic and reviewing its profile before booking. [8]

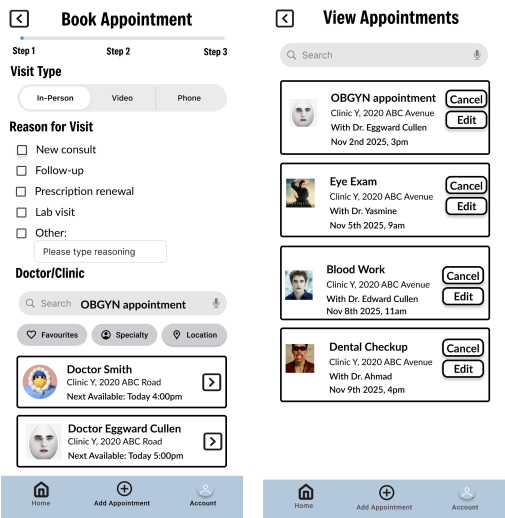


(Fig 5.2a)

(Fig 5.2b)

5.3 Task 2: Appointments Booking and Management (Patients)

Task 2 covers appointment booking and ongoing management from the patient side. Patients can either start from a clinic profile via the “Book Appointment” button or from the bottom navigation bar via “Add Appointment”, supporting both clinic-driven and general booking. In the booking flow (Fig 5.3a), patients select a clinic and doctor, then browse available dates and time slots. The interface maintains key context (i.e. the selected clinic and doctor) visible throughout the process while guiding users step by step. After confirming via the “Accept” button, the appointment is added to the “View Appointments” page (Fig 5.3b), where all upcoming appointments appear as a list of cards with clinic, provider, date, and time. From this overview, patients can also edit an appointment (i.e. change date or time while keeping the same clinic and provider) or cancel it entirely. This task allows patients to book, modify, or cancel appointments as needed. [8]



(Fig 5.3a)

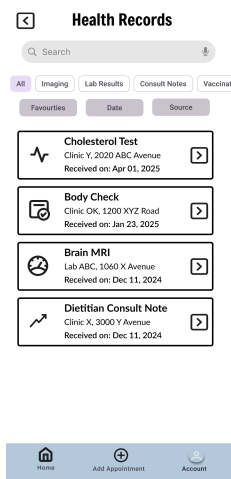
(Fig 5.3b)

5.4 Task 3: Health Records and Prescriptions Management (Patients & Doctors)

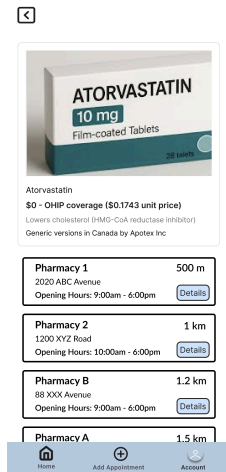
Patient Workflow:

Task 3 supports ongoing care by providing patients with access to their health records and prescriptions through dedicated entry points on the home screen. The “Health Records” page (Fig 5.4a) lists

consultation notes and test results in a searchable, filterable layout (e.g. by type or date). Selecting a record opens a two-level detail view with a concise summary at the top and full test results below, allowing both quick review and deeper reading. The “Prescriptions” page follows a similar structure, but the detailed view (Fig 5.4b) includes drug information and a list of nearby pharmacies for pickup. Clicking a pharmacy opens a “Pharmacy Details” page with more information. This flow enables patients to locate, review, and interpret their records and prescriptions in an organized manner. [8]



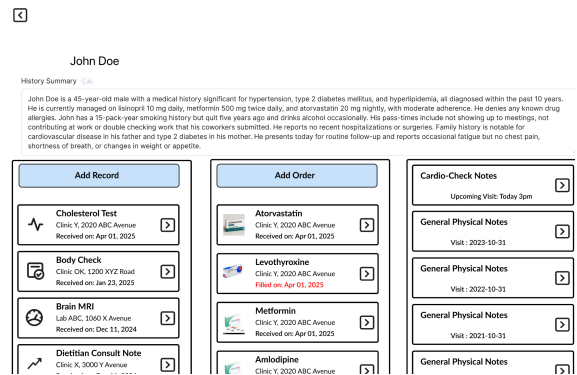
(Fig 5.4a)



(Fig 5.4b)

Doctor Workflow:

On the doctor-facing web dashboard, Task 3 focuses on creating and maintaining patient health data. From the doctor’s home page, clinicians select a patient to open a profile (Fig 5.4c) with a general summary at the top and three columns below. The left column lists health records, where doctors can add, view, and update entries. The middle column lists prescriptions, which can similarly be added or edited to reflect treatment changes. The right column contains a doctor’s notes section for attaching notes to visits or ongoing issues, supporting continuity of care. This flow enables doctors to manage patients’ health records and prescriptions in a structured way that aligns with what patients see on their side. [9]



(Fig 5.4c)

6. Usability Study

6.1 Data-Collection Method:

Our team chose questionnaires as our primary data-collection method due to their overall effectiveness. Questionnaires are a scalable and appropriate tool for evaluating user studies, especially in the user-satisfaction category. Since our prototype includes complex flows (appointments, health records, prescriptions, and clinic browsing), questionnaires allow us to systematically gather both quantitative and qualitative feedback from our two main user groups (healthcare workers vs. patients)[7].

6.2 Study Background:

The usability study was set in a remote environment while using an online Google Form. Participants accessed the prototype flow through Figma and a questionnaire on their own devices (laptop or mobile phone) at their own convenience, without a researcher present. The questionnaire includes the required consent form for privacy and personal information access purposes[10].

6.3 Questionnaire Categories:

We designed the questionnaire to reflect the structure of our prototype (see 5.1), which contains two major user flows: one for healthcare workers (doctors, nurses, healthcare registration workers, etc) and one for patients. Due to the different interactions the two groups have with the system and have distinct priorities, each role is directed to a separate set of feedback questions specifically tailored to their

experience. Our team provided separate links for the two questionnaire sections, ensuring that each participant could easily access and respond to the section that corresponds to their specific role within the system. In addition to role-specific questions(Fig 6.3a), the questionnaire also incorporates sections on personal background and overall satisfaction(Fig 6.3 b). Together, these components divide the survey into four distinct categories, which allows us to gather both customized feedback on each specific user flow and overall perceptions of the system and widen our horizons into user demographics[7].

Physician/Nurse/ healthcare administration Section

For the positions listed above, please visit the link below and answer the following questions in section 3
[Link to Dr View](#)

Adding a new patient record was intuitive

1 2 3 4 5

☆ ☆ ☆ ☆ ☆

(Fig 6.3a)

Satisfaction

what feature do u like the most about our application *

☐ minimalist design

☐ overall usability

☐ record accessibility

☐ search features(clinic)

☐ book/edit/cancel appointments

☐ readability

☐ Other: _____

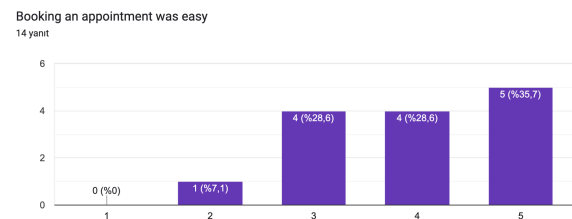
Is there anything you were dissatisfied with while using this app? If yes, please describe

Your answer _____

(Fig 6.3b)

along with several nurses, ensuring feedback from both everyday users and healthcare professionals. Just over half of the participants had previous experience with digital healthcare applications, providing us with a range of perspectives on ease of learning, clarity, and navigation.

Overall, the appointment-related features performed well. Participants rated the ease of booking appointments at 79%(see Fig 6.4a), editing or cancelling appointments at 80%, and the clarity of the clinic list at 79%. These results show that users found the appointment flow intuitive and straightforward. For the Health Records section, performance was slightly higher: the ability to locate records scored 82%, while understanding test results scored 77%. Users generally felt confident navigating these features, though a few noted that some labels and result descriptions could be clearer.

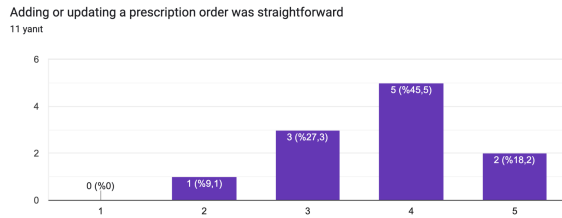


(Fig 6.4a)

Feedback on the doctor and nurse administrative tools was slightly lower, with ratings between 73% and 76% for adding new records (see Fig 6.4b), updating existing ones, and managing prescriptions. Participants appreciated the organizational structure and clear separation of different medical data types. However, they pointed out two recurring issues: some buttons behaved inconsistently, and there was limited confirmation feedback after saving or updating records. This lack of feedback created hesitation and reduced confidence in completing tasks.

6.4 Usability Testing Results:

We conducted a usability evaluation of our healthcare app prototype using a structured questionnaire completed by 15 participants between the ages of 20 and 50[7]. The sample included primarily patients,



(Fig 6.4b)

Across all features, the most common usability problems involved interaction reliability, such as buttons that occasionally did not respond or led to unexpected screens, and inconsistencies in labelling or filter behaviour. These issues did not prevent task completion but occasionally introduced confusion or slowed down user performance.

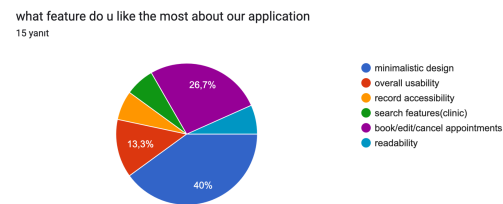
6.5 Overall Results:

When taken together, the usability results indicate that the prototype is already strong in its core design and functionality. Most features scored between 73% and 82%, showing that users were able to navigate major workflows easily. Participants liked the app's clean, minimalistic design, the structure of the interface, and the separation of features such as appointments, medical records, and prescriptions (see Fig 6.5a). These strengths suggest that the basis of the app aligns well with user expectations.

However, to achieve a higher level of usability and reliability, the system requires improvements in several areas. The primary weaknesses identified were button responsiveness, screen-to-screen consistency, and lack of confirmation feedback. Addressing these issues will enhance user trust and reduce errors in both patient and clinician workflows. In addition, refining labels, improving filter behaviour, and increasing the clarity of test result displays will make the user experience better.

In summary, the usability testing demonstrates that our prototype successfully supports the key tasks expected from both patients and healthcare providers. While the overall experience is positive, refinements in interaction reliability and interface consistency will

be essential for preparing the system for real-world use.



(Fig 6.5a)

7. Limitations

Although the study provided valuable insights into how users interact with the system, several limitations should be acknowledged to better contextualize the findings. These limitations highlight areas where the study's scope, participant diversity, and testing conditions could be broadened to gain a more complete understanding of user needs and real-world performance.

7.1 Limited Range of Participants

Most participants were between 20–50 years old and were primarily patients. There were very few older adults, beginners with technology, or people with disabilities, groups that often face the greatest challenges in healthcare apps. As a result, the findings may not fully represent their needs.

7.2 Mid-Fidelity Prototype Constraints

The prototype tested was not a fully built product. Several interactions, such as interface responses, buttons, and feedback messages, were only partially functional. Issues like misfiring buttons or inconsistent behaviour may stem from prototype limitations rather than the design itself.

7.3 Short, Task-Based Testing

Participants completed only a few tasks in short sessions. This does not reflect long-term behaviour such as reviewing extended medical records, booking multiple appointments, or managing ongoing prescriptions.

7.4 No Formal Accessibility Testing

Although inclusivity was a design goal, formal accessibility testing was not conducted. Screen readers, large-text modes, and high-contrast settings were not evaluated. Users with visual, motor, or cognitive impairments were not included, limiting understanding of actual accessibility performance.

7.5 Limited Device and Real-World Coverage

Testing took place mostly on desktop/mobile devices in controlled environments. The system was not evaluated on tablets or in real-world scenarios such as clinic waiting rooms, nighttime usage, or multitasking situations.

7.6 Small Sample of Doctors and Nurses

Only a few healthcare professionals participated. Their workflows differ significantly from patients, so the limited sample size reduces confidence in evaluating the provider-facing interface.

8. Future Work & Improvements

Building on the insights and constraints identified in the study, several opportunities emerge for strengthening the design and expanding its impact. The following phases outline a roadmap for improving usability, accessibility, and long-term adoption, as well as supporting broader, more realistic use cases in healthcare settings.

8.1 Phase 1: Short-Term Improvements (0–3 Months)

In the near term, the main priority is to improve how reliably the interface responds to people's actions. Several users experienced misfiring buttons or inconsistent behaviour, so refining the touch areas, standardizing button placement and labels, and making the overall interaction feel more predictable will help build trust. The filtering and search options also need to work more smoothly; fixing the underlying logic and adding quick-loading indicators can make these features feel much more responsive. Another area that needs attention is system feedback. Users often weren't sure if their updates were saved, so adding clear confirmations and more helpful error messages will make the experience feel more reassuring. Finally, some labels in the interface were

confusing or too vague, so simplifying terminology and adding small contextual explanations where needed will help reduce ambiguity.

8.2 Phase 2: Medium-Term Improvements (3–6 Months)

As the system matures, expanding accessibility and supporting clinical workflows more effectively should become the focus. Full accessibility support, such as improving screen-reader compatibility, offering high-contrast and large-text modes, and adding optional haptic feedback, will help make the app usable by a wider range of people, including those with visual or motor challenges. On the clinical side, doctors and nurses expressed that certain tasks felt effortful or uncertain. To address this, the interface can provide clearer summaries after edits, autosave long entries so progress isn't lost, and introduce quick templates for commonly repeated medical notes. These improvements will help reduce the cognitive load for busy clinicians and make their everyday interactions with the system more efficient.

8.3 Phase 3: Long-Term Enhancements (6–12 Months)

Over the longer term, the design should be scaled and refined for real-world use across different devices and environments. Since the study didn't include tablet users or testing across operating systems, the interface should be optimized for larger screens and checked for consistent behaviour on iOS, Android, and Windows. It may also be helpful to adapt the layout depending on whether the user is a patient or a healthcare provider, since their needs are quite different. Alongside this, a longer observational study, spanning several months, would offer a much clearer picture of how people actually use the app in daily life. This would help uncover patterns in patient engagement, how often medical records are revisited, and whether the system influences appointment behaviour over time.

8.4 Phase 4: Broader Future Work (12+ Months)

Looking further ahead, there is room to explore how the system could support a broader range of users and clinical situations. One important direction is

multi-language support, especially for people with lower literacy skills. Studying how translated interfaces affect comprehension, booking accuracy, and user confidence would help ensure the system is inclusive in practice, not just in intention. There is also strong potential in adding AI-driven features, such as smart appointment suggestions, automatic summaries of complex medical reports, or predictive reminders for medications. These tools could be particularly helpful for older adults and people managing chronic conditions. Finally, testing the system directly inside clinics, during busy hours, under real pressure, would reveal workflow interruptions, multitasking needs, and other practical challenges that are difficult to simulate in controlled lab sessions. Understanding these factors will be essential for designing a system that truly fits into everyday clinical practice.

9. Conclusion

This project demonstrated how user-centred design methods and HCI principles can meaningfully address the fragmentation present in current digital health systems. Through surveys, interviews, and iterative prototyping, we identified that both patients and clinicians struggle with dispersed information, inconsistent workflows, and limited accessibility across existing platforms. Our interactive prototype and usability study showed that a centralized, well-structured system can significantly improve task clarity, navigation, and overall user experience. At the same time, our findings highlighted areas requiring refinement, particularly interaction reliability, consistent feedback, and broader accessibility support. While the work remains exploratory, the results offer a clear direction for building more integrated, equitable, and user-friendly digital health tools, setting the groundwork for future development and evaluation.

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